RaceDay API Endpoint Plan

Part 1, Section B

This table lists every endpoint the RaceDay REST API will expose in Part 2.

It covers Authentication, User Profile, Events, Categories, Event Enrolments,

and Results, as required. All routes are prefixed with `/api/`.

1. Authentication

HTTP Method, Route, Description, Role Required, Request Body, Expected Response,

(POST)/api/auth/register. Creates a new user account as either an Organiser or a Participant. None (public) | { firstName, lastName, email, password, role } 201 Created – user created; 400 Bad Request – validation error; 409 Conflict – email already registered

| POST | /api/auth/login | Authenticates a user and returns a JWT access token. None (public) | { email, password } 200 OK – token + role returned; 401 Unauthorized – invalid credentials

2. User Profile

| HTTP Method | Route | Description | Role Required | Request Body | Expected Response

| GET | /api/users/me | Returns the profile of the currently logged-in user. Any (logged in) | None | 200 OK – user profile object; 401 Unauthorized .

| PUT | /api/users/me Updates the profile details of the logged-in user. Any (logged in) { firstName, lastName, contactNumber } 200 OK – updated profile; 400 Bad Request

| DELETE | /api/users/me Deletes the logged-in user's account. Any (logged in) None 204 No Content; 404 Not Found

3. Events

| HTTP Method | Route | Description | Role Required | Request Body | Expected Response

| GET | /api/events | Lists all upcoming events. Supports optional filtering by location/date. None (public) None 200 OK – array of events

| GET | /api/events/{id} Returns the full detail of a single event, including its categories. None (public) None 200 OK – event object; 404 Not Found

| POST | /api/events Creates a new event. Organiser { eventName, eventDate, location, description } 201 Created – new event; 400 Bad Request

| PUT | /api/events/{id} | Updates an existing event owned by the logged-in Organiser. | Organiser | `{ eventName, eventDate, location, description }` | 200 OK – updated event; 403 Forbidden – not the owner; 404 Not Found |

| DELETE | /api/events/{id} Deletes an event owned by the logged-in Organiser. Organiser None 204 No Content; 403 Forbidden; 404 Not Found

4. Categories

| HTTP Method | Route | Description | Role Required | Request Body | Expected Response |

| GET | /api/events/{eventId}/categories | Lists all categories for a specific event. | None (public) | None | 200 OK – array of categories; 404 Not Found |

| POST | /api/events/{eventId}/categories | Adds a new category to an event. | Organiser | `{ categoryName, distanceKm, maxParticipants, entryFee }` | 201 Created – new category; 403 Forbidden; 404 Not Found |

| PUT | /api/categories/{id} | Updates an existing category. | Organiser | `{ categoryName, distanceKm, maxParticipants, entryFee }` | 200 OK – updated category; 403 Forbidden; 404 Not Found |

| DELETE | /api/categories/{id} | Removes a category from an event. | Organiser | None | 204 No Content; 403 Forbidden; 404 Not Found |

5. Event Enrolment

| HTTP Method | Route | Description | Role Required | Request Body | Expected Response

| POST | /api/categories/{categoryId}/enrol Enrols the logged-in Participant into a category.Participant { } (participant identified from token) 201 Created – enrolment record; 404 Not Found – category does not exist; 409 Conflict – already enrolled or category full

| GET | /api/enrolments/me Lists all enrolments belonging to the logged-in Participant. Participant None 200 OK – array of enrolments

| GET | /api/events/{eventId}/enrolments Lists all enrolments for an event (for the Organiser managing it). Organiser None 200 OK – array of enrolments; 403 Forbidden; 404 Not Found

| DELETE | /api/enrolments/{id} Cancels an enrolment. Participant None 204 No Content; 403 Forbidden – not the owner; 404 Not Found

6. Results

| HTTP Method | Route | Description | Role Required | Request Body | Expected Response |

| POST | /api/enrolments/{enrolmentId}/results Captures a finish result for a Participant's enrolment. Organiser { finishTime, positionOverall, positionCategory } 201 Created – result record; 403 Forbidden; 404 Not Found; 409 Conflict – result already captured

| GET | /api/results/me | Returns the logged-in Participant's personal result history. Participant None 200 OK – array of results

| GET | /api/events/{eventId}/results Returns all results for an event. None (public) None 200 OK – array of results; 404 Not Found

| PUT | /api/results/{id} Corrects a previously captured result. Organiser { finishTime, positionOverall, positionCategory } 200 OK – updated result; 403 Forbidden; 404 Not Found

Design notes

-**Role enforcement:** happens via the JWT claims issued at login — the `role` claim is checked in an `[Authorize(Roles = "Organiser")]` (or equivalent) attribute on every protected endpoint in Part 2.

- Routes that modify a resource (`PUT`/`DELETE`) always confirm the logged-in user actually **owns** that resource (e.g. an Organiser can only edit their own events) before allowing the change — this is why several endpoints list a `403 Forbidden` response in addition to `404 Not Found`.

- This plan matches the ERD in `RaceDay_ERD.png` exactly: one endpoint group per entity, plus the `EventEnrolments` join endpoints that resolve the many-to-many relationship between Participants and Categories.